

Prediction Uncertainty

Confidence Intervals for Prediction and Prediction Intervals

Last week

We learned 2 important topics

- inference after model selection (double-dipping problem)
- regularized methods

Last week in a nutshell

(you still need to know everything covered in previous classes)

The post-inference problem

- we run a simulation study and showed that the type I error rate got seriously inflated when the same data is used to select variables and conduct inference.

This problem is known as the post-inference problem (aka “doble dipping”)

- the inference results computed on reused data are not valid (as seen in the first part of the worksheet).
- **A solution:** if we split the data, we can use one part to select and the other part to estimate and build tests.
- we showed with a simulation study that the type I error rate was controlled with this strategy.

Variable Selection Methods

~~Scroll down to see full content~~

- **Stepwise Selection Algorithms** add (or remove) variables sequentially. They compare models of the same size using RSS and models of different sizes using appropriate metrics (e.g., $\text{adj}R^2$, AIC, BIC, C_p)
 - different algorithms depending on the type of covariates (e.g., categorical with multiple levels)
- **Regularized Methods** smoothly shrink the estimated coefficients. They minimize the RSS but subject to a bound on the size of the coefficients
 - **Ridge** uses an L_2 –norm to measure the size of the coefficients

$$\|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2$$

- **Lasso** uses an L_1 –norm to measure the size of the coefficients

$$\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$$

MINIMIZATION SUBJECT TO A BOUND ON THE SIZE OF THE COEFFICIENTS

222 6. Linear Model Selection and Regularization

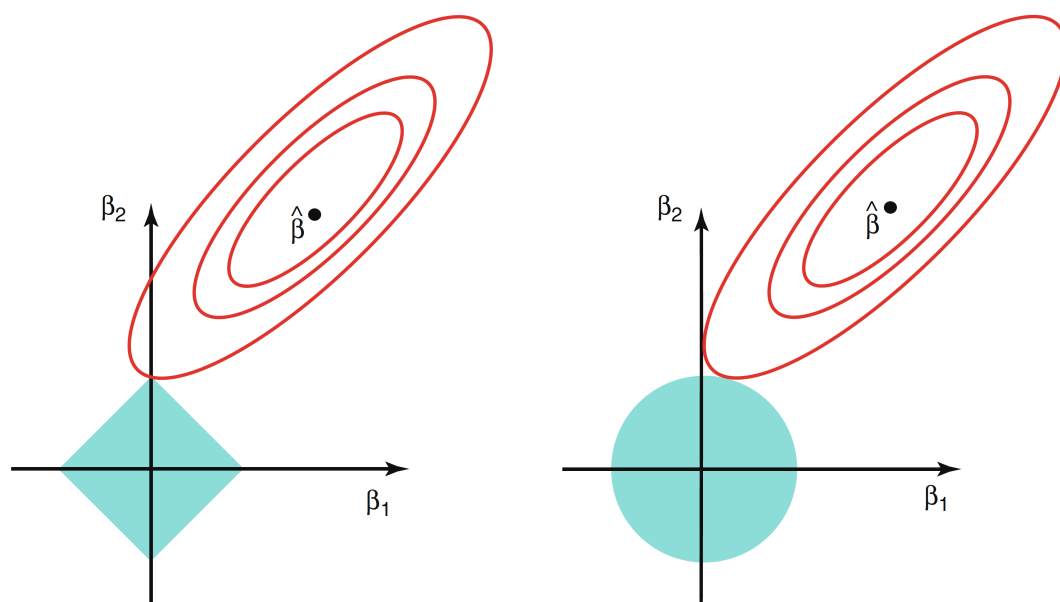
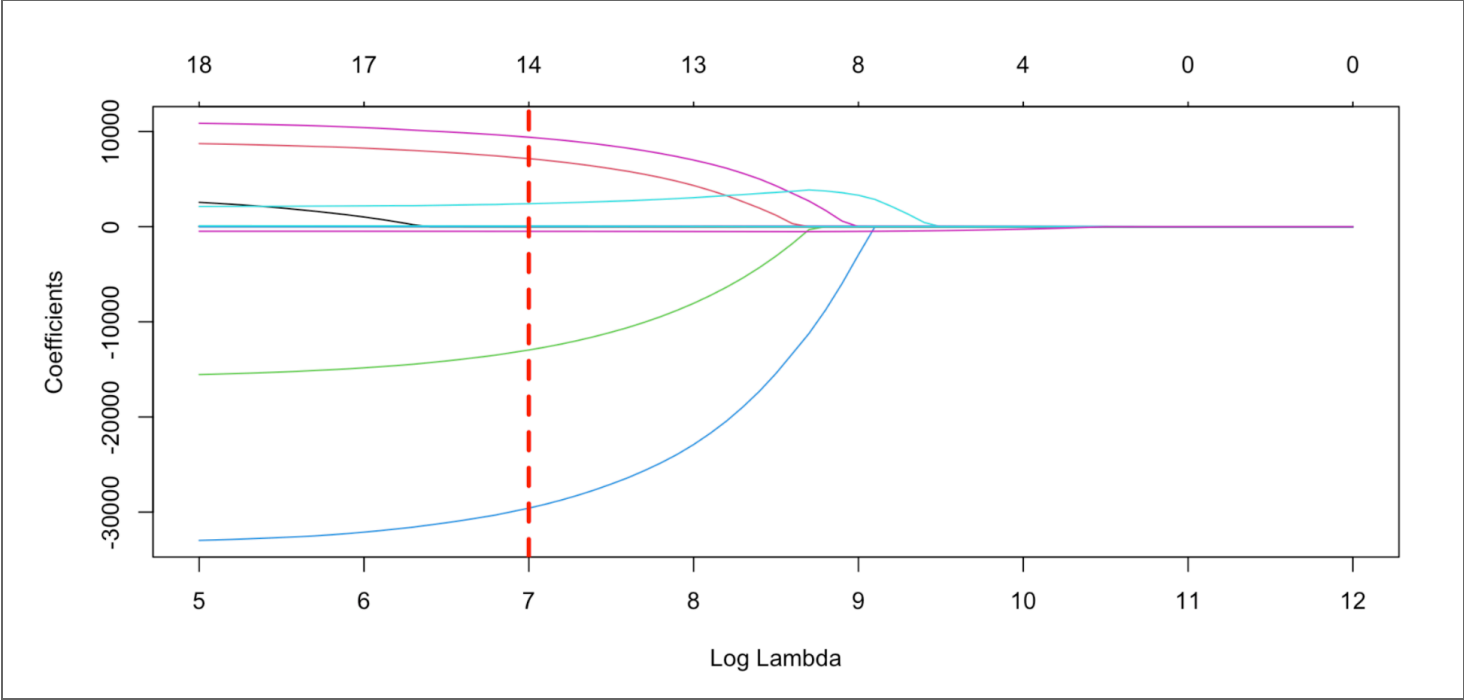


FIGURE 6.7. Contours of the error and constraint functions for the lasso (left) and ridge regression (right). The solid blue areas are the constraint regions, $|\beta_1| + |\beta_2| \leq s$ and $\beta_1^2 + \beta_2^2 \leq s$, while the red ellipses are the contours of the RSS.

Coefficients are shrunk as the level of penalty increases



iClicker

The value of the estimated coefficient at the dotted line correspond to the least squared estimated coefficients (OLS) calculated with `lm()` function

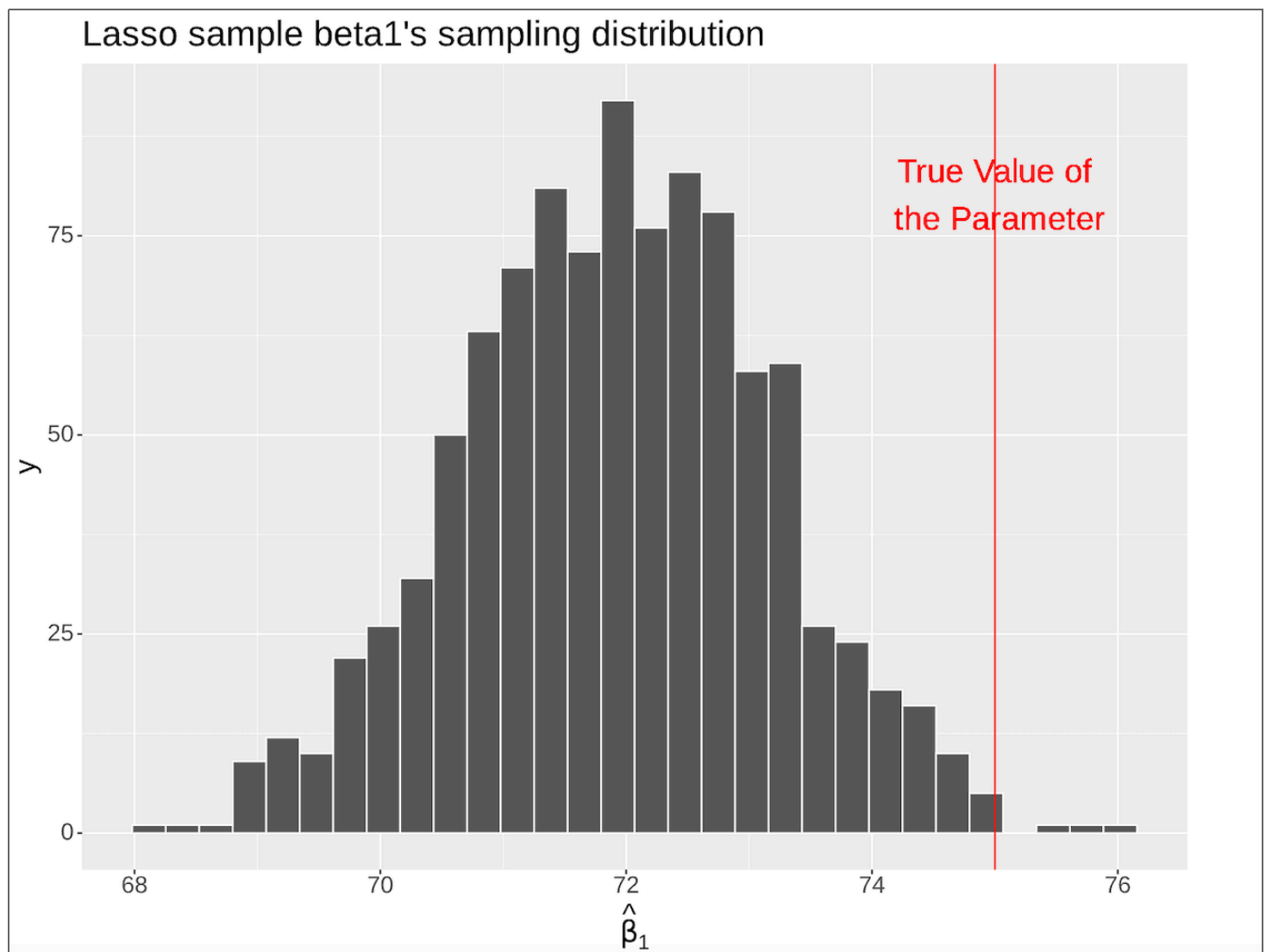
A. TRUE

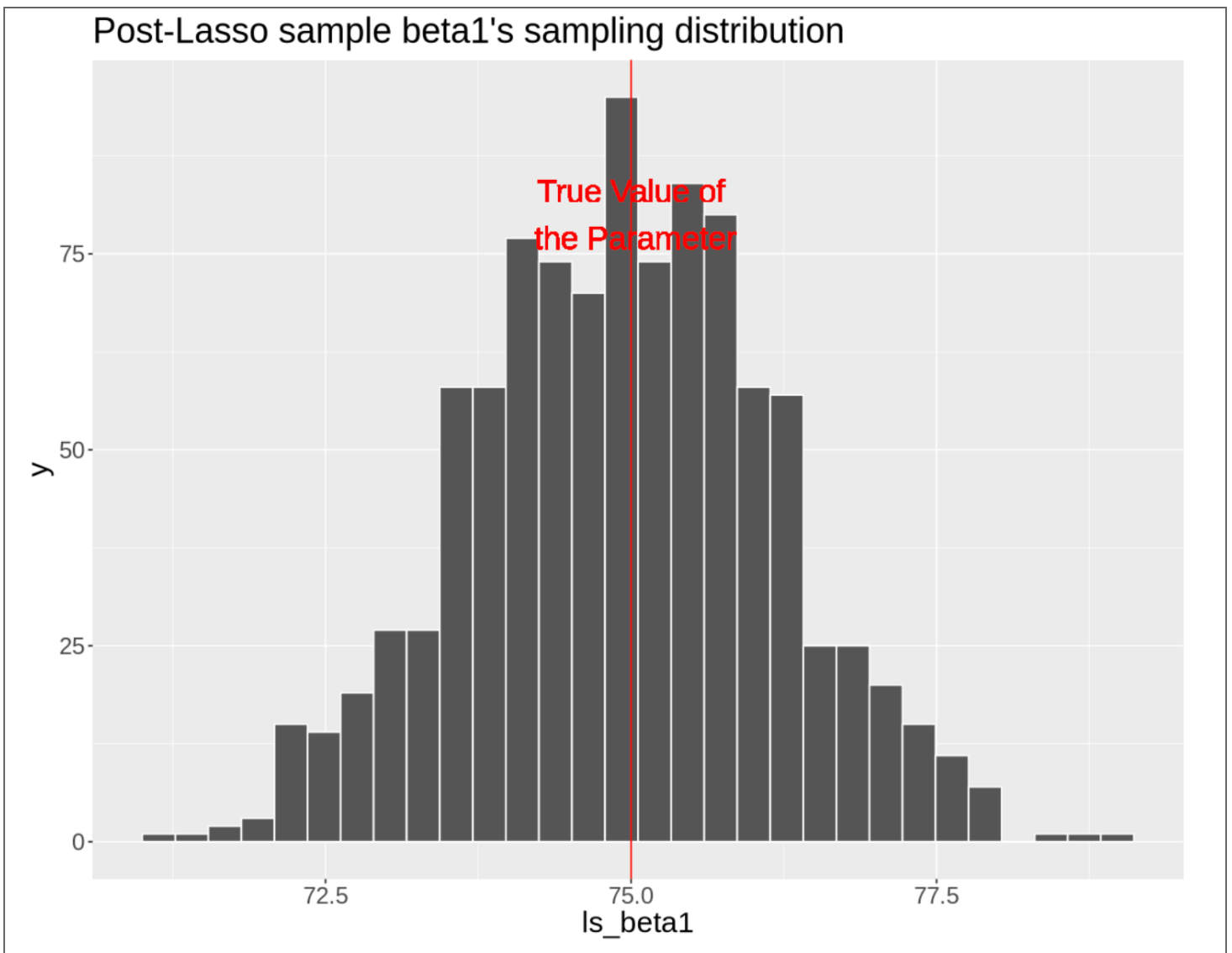
→ B. FALSE

- The bound (or equivalently, level of penalty λ) is usually selected by cross-validation aiming to minimize the MSE
- With enough penalization, LASSO shrinks all estimated coefficients to zero. A way to select variables!
- Ridge will never reach a value of zero and thus *can not* be used to select variables.
- This “shrinkage” process biases the estimated coefficients to favor prediction performance.

Debiasing LASSO

Scroll down to see full content





IClicker: postLASSO ESTIMATOR IS A LS ESTIMATOR ON THE SELECTED VARIABLES, THUS IT'S UNBIASED

→ A: TRUE

B: FALSE

- We can fit a LS estimator using *only* the variables selected by LASSO. This estimator is known as **postLASSO** and is **unbiased!**

- If we want to make inference with postLASSO, we need to split the data!

Prediction Uncertainty

Today

Predictions using the estimated models depend on the sample used, thus it is subject to sample-to-sample variation
We'll take into account the associated uncertainty using:

- confidence intervals for prediction (CIP)
- prediction intervals (PI)

What is the difference between these??

Predictions are random variables

We learned how to **estimate** a model using a **random sample**.

- the estimated model can be used to **predict** the response of new observations.
- the predictions depend on the random sample used! In other words, the predictions are functions of the a random sample.

Which means that **the predictors are also random variables**

Prediction uncertainty

A different sample would result in a different estimated model and, thus, different predictions!

- the sample-to-sample variation in the estimated coefficients translates into variation in the predictions
- we can account and report this **uncertainty** of the predictions using *intervals*

We'll focused on predictions using MLR. Extensions to GLMs are not easy.

Dataset

In worksheet_09, you'll work with the 2015 Property Tax Assessment from Strathcona County Dataset

The dataset contains data on property tax-assessed values in Strathcona County in 2014.

The Property Tax Assessment dataset has information about 27699 properties

```

                                the_geom TAX_YEAR  ROLL_NUM
1 POINT (-113.27665515325147 53.54627975402534)    2015 -2147483648
      ADDRESS YEAR_BUILT  ASSESSCLAS          BLDG_DESC BLDG_METRE
1 447 CASCADE CR      2010 Residential 2 Storey & Basement      201
  BLDG_FEET GARAGE FIREPLACE BASEMENT BSMTDEVL ASSESSMENT LATITUDE
LONGITUDE
1      2162      Y      Y      Y      N      580000 53.54616
-113.2769
  assess_val
1      580
```

To learn concepts from this topic, we'll treat the entire dataset as a finite population and draw a random sample from it to estimate and predict population quantities.

```
set.seed(561) # DO NOT CHANGE THIS.  
  
properties_sample <-  
  properties_data %>%  
  slice_sample(n = 100, replace = FALSE)
```

We do NOT do this in practice. If we know the population we don't need to take a sample!

Population vs Estimated SLR

- You can **estimate** the population SLR using the sample `properties_sample`
- You can compute the population coefficients of the SLR using all properties in `properties_data`

Again, in practice you would not be able to *compute* the population coefficients

```
lm_pop <- lm(assess_val ~ BLDG_METRE, properties_data)
lm_sample <- lm(assess_val ~ BLDG_METRE, properties_sample)

tidy(lm_pop) %>%
  mutate_if(is.numeric, round, 3)
```

```
# A tibble: 2 × 5
  term          estimate std.error statistic p.value
<chr>         <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   98.4       2.04      48.1     0
2 BLDG_METRE    2.56      0.012    207.     0
```

```
tidy(lm_sample) %>%
  mutate_if(is.numeric, round, 3)
```

```
# A tibble: 2 × 5
  term          estimate std.error statistic p.value
<chr>         <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   34.5      38.1      0.905   0.367
2 BLDG_METRE    3.07      0.228     13.4     0
```

Prediction of assessed value

Using linear regression, the assessed value of a random house in Strathcona can be modelled as the average assessed value of a house with similar characteristics plus some random error.

Mathematically,

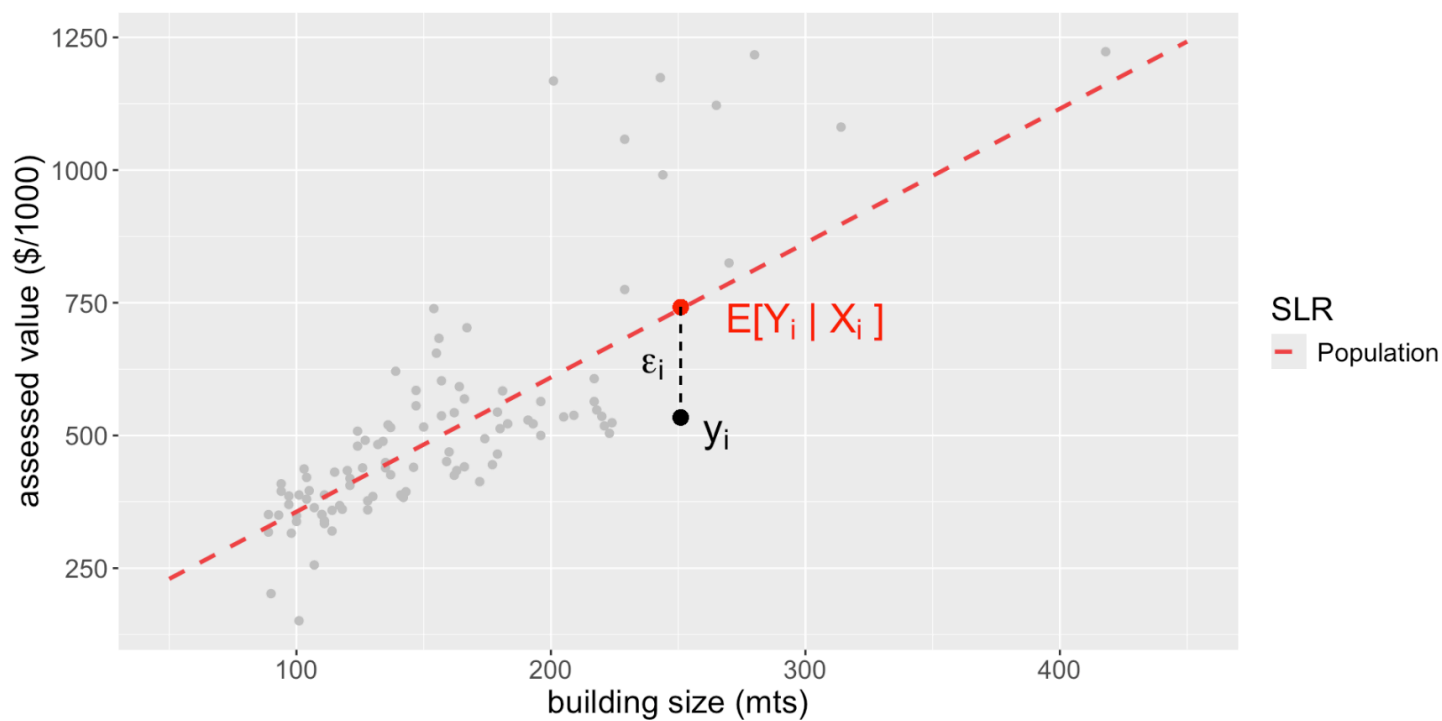
$$Y_i = E[Y_i|X_i] + \varepsilon_i$$

The error, ε_i , is used to model fluctuations around the average population value of residencies of the same size.

If we assume that the conditional expectation is linear, then:

$$E[Y_i | X_i] = \beta_0 + \beta_1 X_i$$

which is the population regression line!

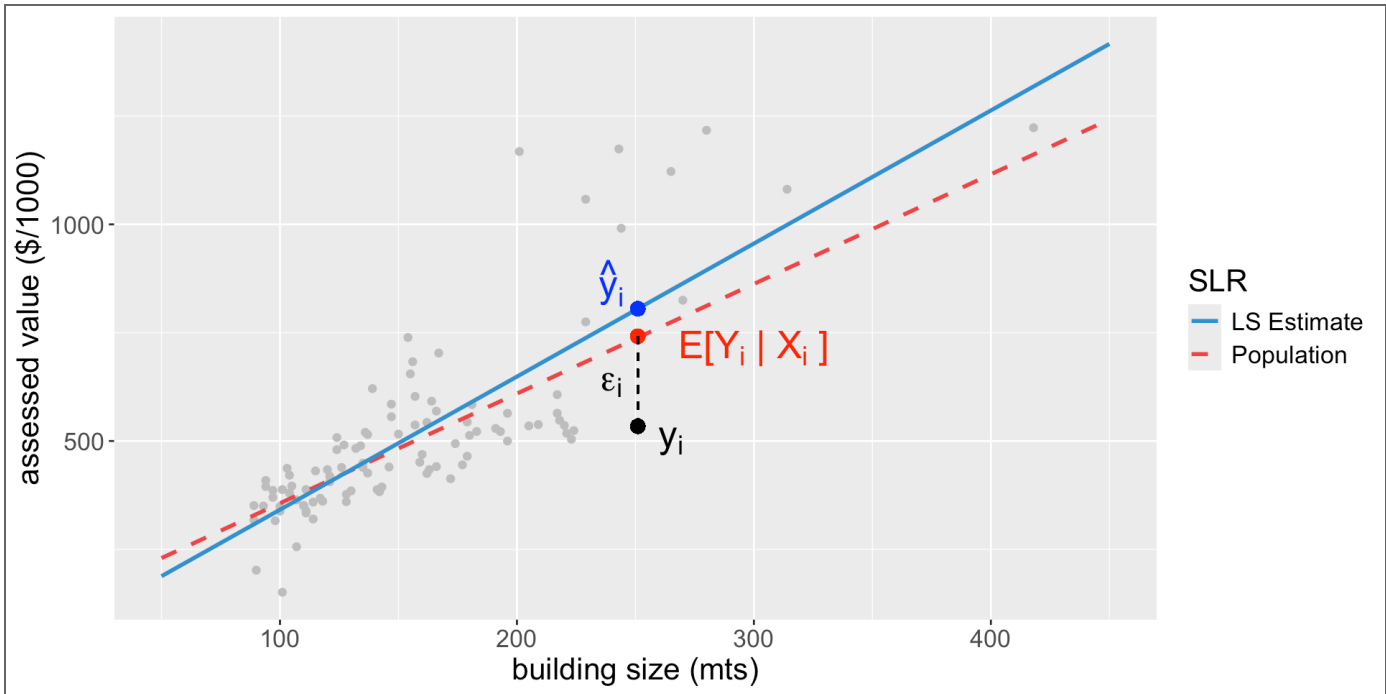


Estimated SLR

We use the random sample to **estimate** the regression line. The **estimated** SLR can be used to **predict** the value of any house in the county.

The prediction of the i -th property is given by:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$



- in practice, ε_i will always be unknown since, $E(Y_i | X_i)$ is unknown
- the residuals r_i can be computed from the observed and predicted responses

Intervals to describe uncertainty

What do we want to predict with $\hat{Y}_i$? In general, we are interested in predicting either:

- the *average* assessed value of a house of *this size*:
 $E[Y_i | X_i]$
- the *actual* value of a house of *this size*: Y_i (knowing its size X_i)

We predict both with *uncertainty*!

iClicker question

Which one do you think is more difficult to predict?
Why?

- A. It is more difficult to predict the *average* assessed value
- > B. It is more difficult to predict the *actual* assessed value
- C. They are equally uncertain and difficult to predict

Confidence Intervals for Prediction

CIP are used when we want to predict $E[Y_i | X_i]$ (conditional expectation)!

The predicted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ approximates, with uncertainty, the population average value

$$E[Y_i | X_i] = \beta_0 + \beta_1 X_i$$

if we take a different sample, we get different estimates (i.e., different blue lines) and, consequently, different predictions

The only source of variation here is the sample-to-sample variation

A **95% confidence interval for prediction** is a range that has a 95% probability of capturing the **population average** value of a house with size X_i

Once we have estimated and predicted values, the range is non-random so we use the word **confidence** (instead of “probability”) since nothing else is random!

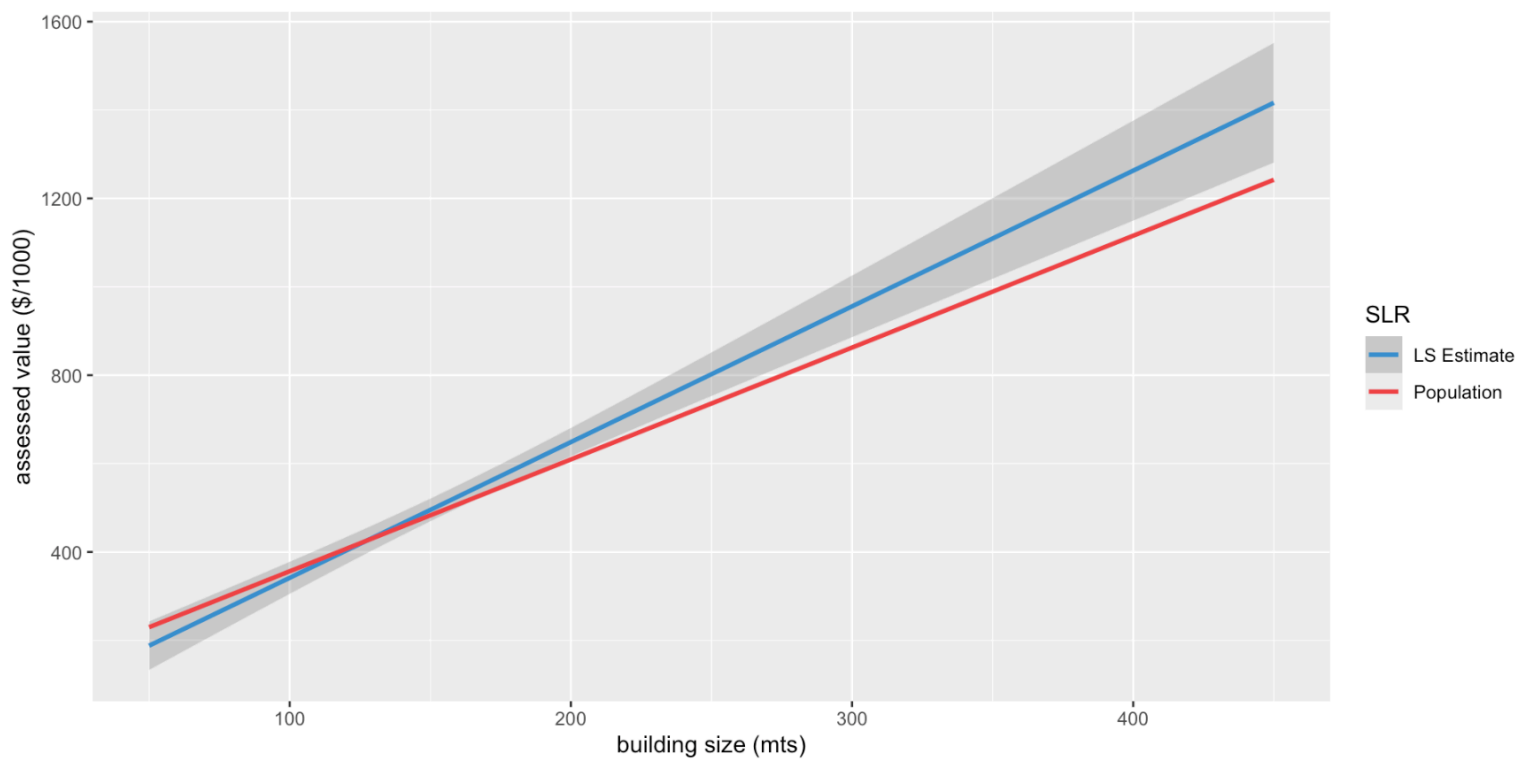
Intervals for conditional expectations

```
properties_cip <-  
  properties_sample %>%  
  select(assess_val, BLDG_METRE) %>%  
  cbind(predict(lm_sample, interval="confidence", se.fit=TRUE)$fit)
```

assess_val	BLDG_METRE	fit	lwr	upr
536	220	710.071	671.944	748.198
370	97	332.341	295.214	369.467
318	89	307.773	267.914	347.632

Interpretation for row 1:

With 95% confidence, the *average value* of a house of size **220 mts** is between \$671, 944 and \$748, 198 (rounded)



Prediction Intervals

PI are used when we want to predict the actual response of a new observation Y_i !

The predicted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ also approximates, with uncertainty, the actual response $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$.

However, now the uncertainty comes from the estimation (sample-to-sample variability) and the error term that generates the data, **two sources of uncertainty**!!

Two sources of uncertainty:

- **uncertainty 1:** because the estimated value $\hat{\beta}_0 + \hat{\beta}_1 X_i$ *approximates* the average (population) value $\beta_0 + \beta_1 X_i$.
- **uncertainty 2:** because the actual observation Y_i differs from the population average value by an error ε_i

PIs are centered at the fitted value $\hat{Y}_i$ as well, but they are **wider** than the CIP to account for the extra source of uncertainty

A 95% prediction interval is a range within which a new value of a house of this size is expected to fall with a specified probability (e.g., 95%).

This time, the aim is to predict the actual response, which is a random variable, thus we interpret the interval in terms of “probability”.

```
properties_pi <-
  properties_sample %>%
  select(assess_val, BLDG_METRE) %>%
  cbind(predict(lm_sample, interval="prediction"))
```

assess_val	BLDG_METRE	fit	lwr	upr
536	220	710.071	454.519	965.622
370	97	332.341	76.937	587.745
318	89	307.773	51.957	563.588

Interpretation for row 1:

With 95% probability, the *value of a house of size 220 mts* is between \$454, 519 and \$965, 622 (rounded).

iClicker question

Which intervals are wider?

- A. CIP are wider than PI since predicting an *average* value is less uncertain
- B. CIP are wider than PI since predicting an *average* value is more uncertain
- > C. PI are wider than CIP since predicting an *actual* value is more uncertain
- D. PI are wider than CIP since predicting an *actual* value is less uncertain

Conclusions

- Confidence intervals for prediction account for the uncertainty given by the estimated LR to predict the conditional expectation of the response
- Prediction intervals account for the uncertainty given by the estimated LR to predict the actual response, i.e, the conditional expectation of the response *plus* the error that generates the data!
- PIs are wider than CIPs; both are centered at the fitted value!

$$Y_i = E[Y_i | X_{i1}, \dots, X_{ip}] + \varepsilon_i$$



model assumption

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip} + \varepsilon_i$$

which one are you
predicting??



$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \dots + \hat{\beta}_p X_{ip}$$

prediction

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Speaker notes